Simple RPC Library (SRPC)

SRPC is an experimental RPC framework that uses Python interfaces as its Interface Definition Language. Define a service contract once, generate client and server stubs, implement the service, and invoke remote procedures.

Features

- Python as IDL
- Automatic Stub generation
- TLS support

> [!WARNING] Currently, the framework only supports Python language.

Table of Contents

Server side directory structure

```
```

With:

- is the root directory of the service.

Define service contract

```
```

Implement service

```
```

Generate Server Stub

The tool to generate the stubs is . Use it as below:

With:

- the path of the service contract
- specify the target Stub(CLIENT/SERVER)
- Specify the language in which the stub should be generated.

Inside directory run:

It will generate the file

Implement the Server

With: means do not use TLS means the server will use the localhost as IP address

Run the Server

Inside directory run:

You should see something like this:

> [!NOTE] By standard, SRPC uses port 5000.

Client side directory Structure

> [!NOTE] The client and server must use the same service contract. In this example, the contract is copied into both projects.

Generate Client Stub

Inside the directory run:

It will generate the file

Implement the Client

Run the Client

Inside the directory run:

```
```

Naming conventions

Component	Required pattern	Example
Service package	lowercase service name	
Service contract/interface		
Implementation script		
Interface class		
Implementation class		

> [!WARNING] SRPC currently relies on naming conventions for automatic discovery and stub generation.

Important Technical Details

Under the hood, SRPC uses:

- A framed application protocol with fixed-size header
- MessagePack-based serialization(SRPC exchanges messages as byte arrays)
- Generic error communication based on error codes
- One connection and one handler thread are used for each request.

> [!WARNING] No performance tests or rigorous security analysis have been conducted. Do not use this in production environments.

> [!NOTE] Technical documentation and release notes for version v4.x.x in progress...

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